

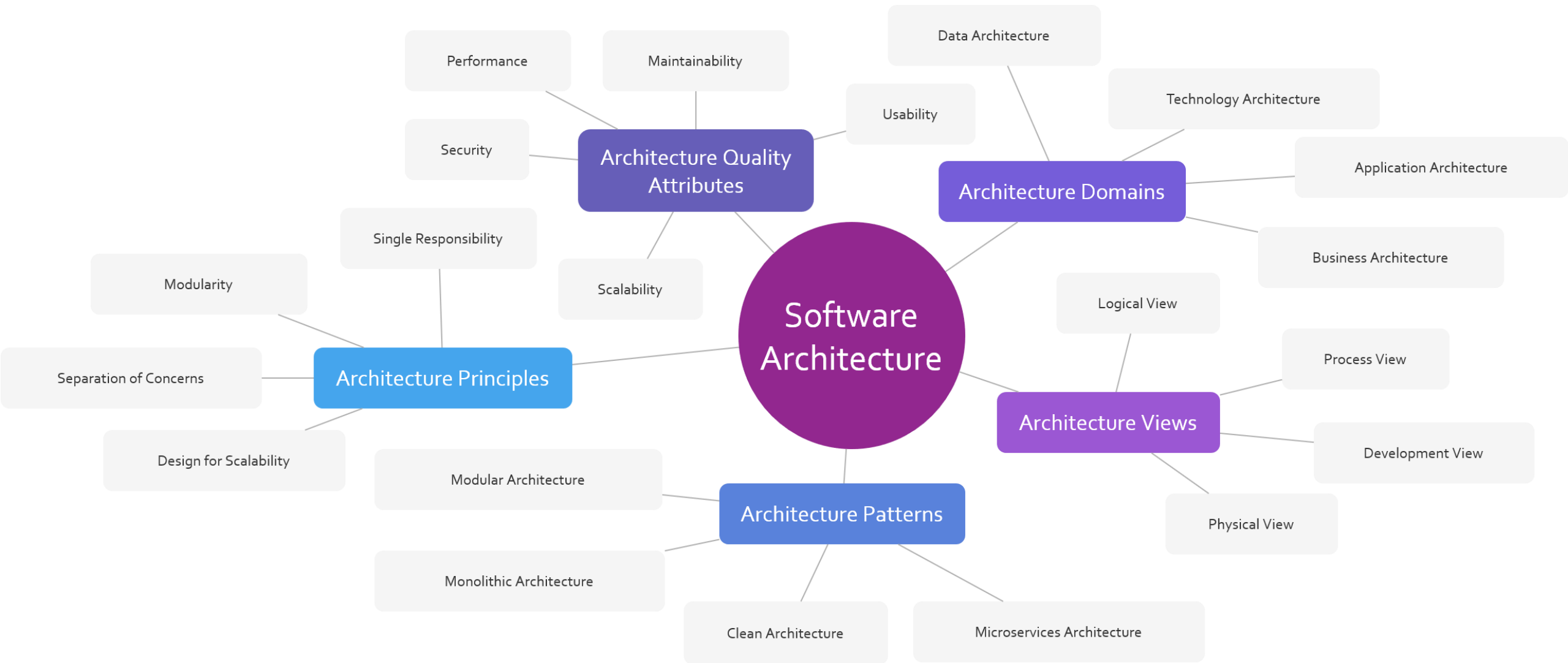
Overview of

Software Architecture, Implementation and Delivery

In Software Development Process

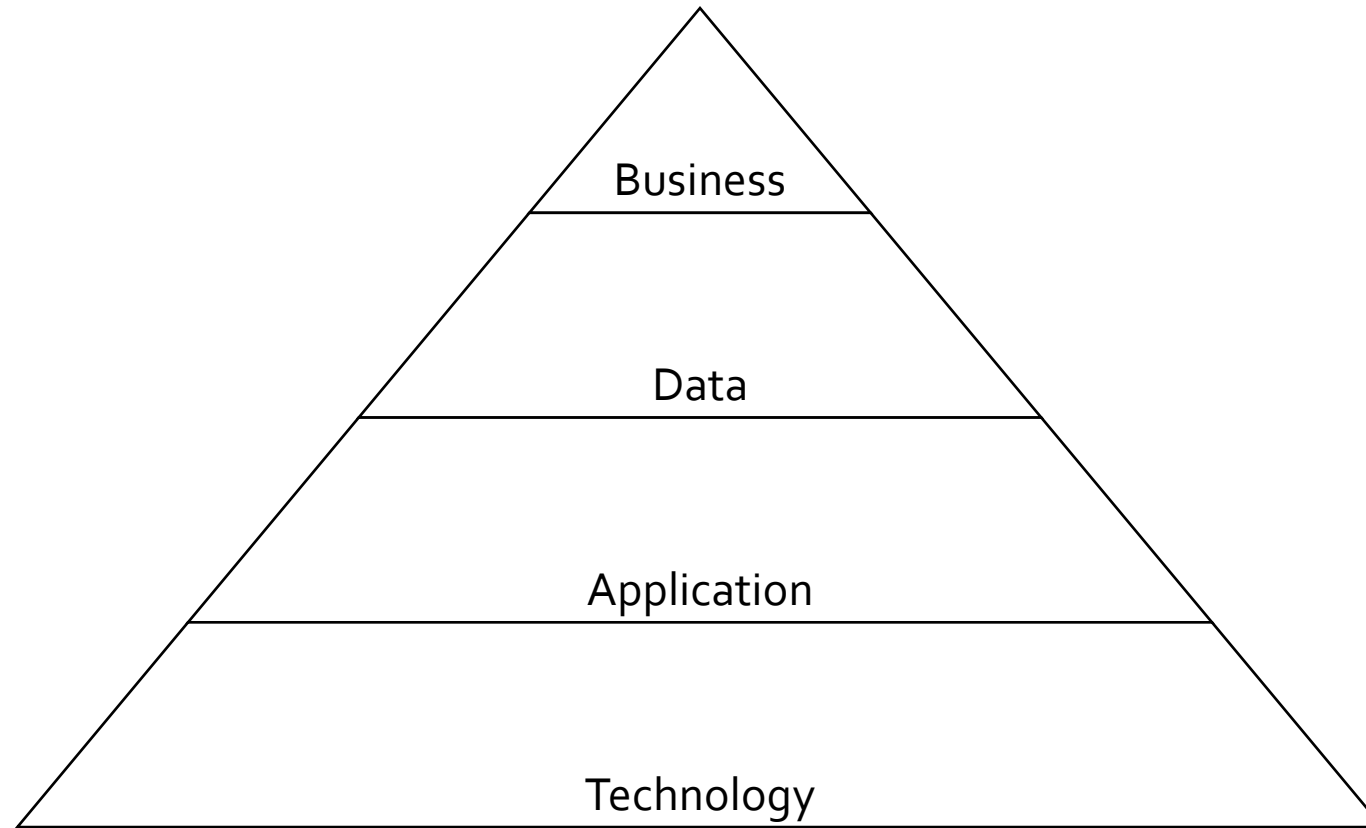
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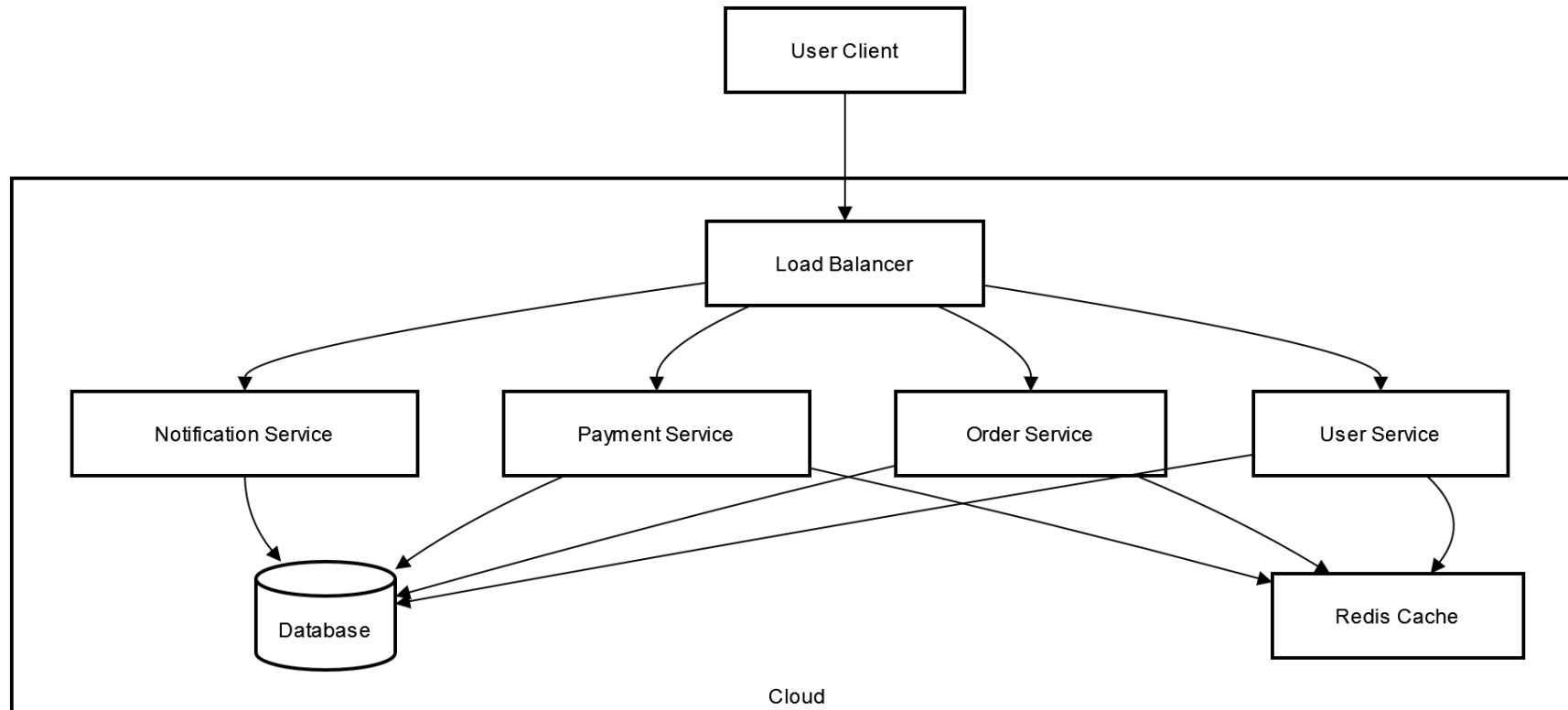
High-level blueprint of a software system that defines its structure, behavior, and key design decisions.

1. Architecture Domains



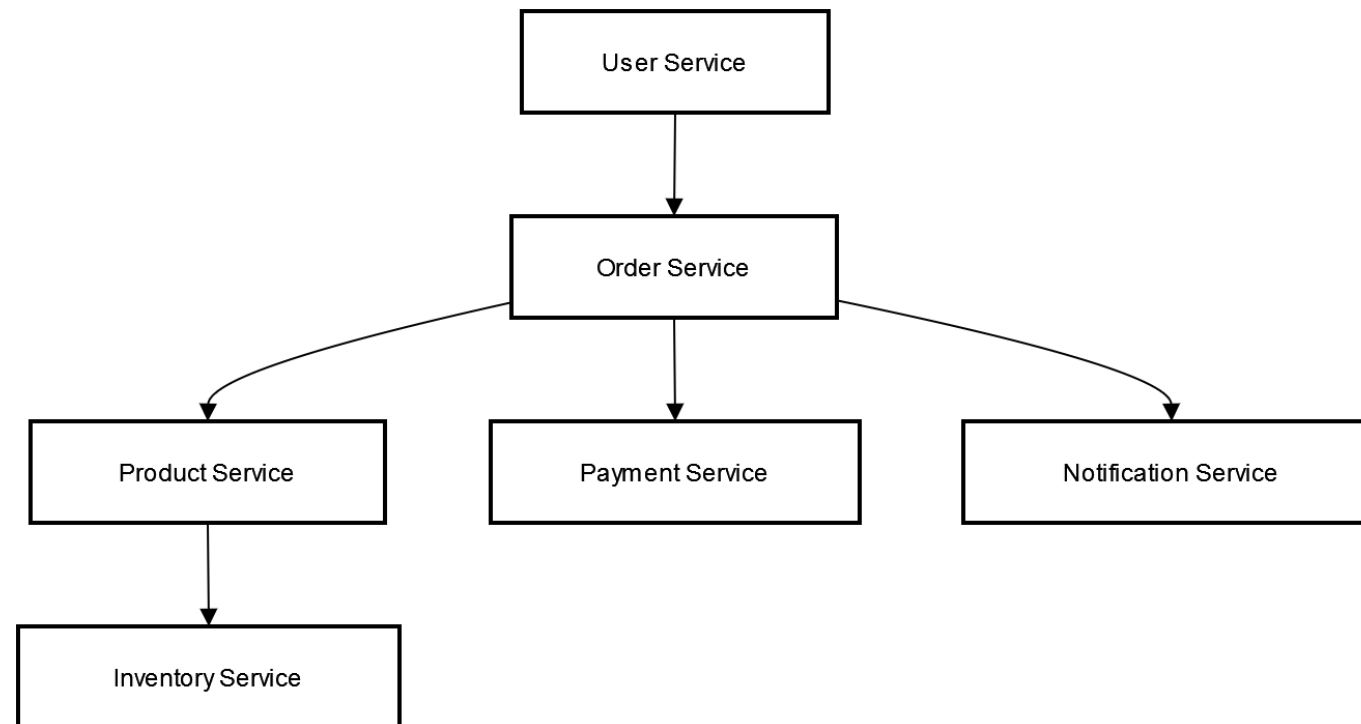
2.1 Architecture Views – Physical View

Describes how the software components are deployed on physical infrastructure or cloud services. It shows the relationship between software components and hardware nodes.



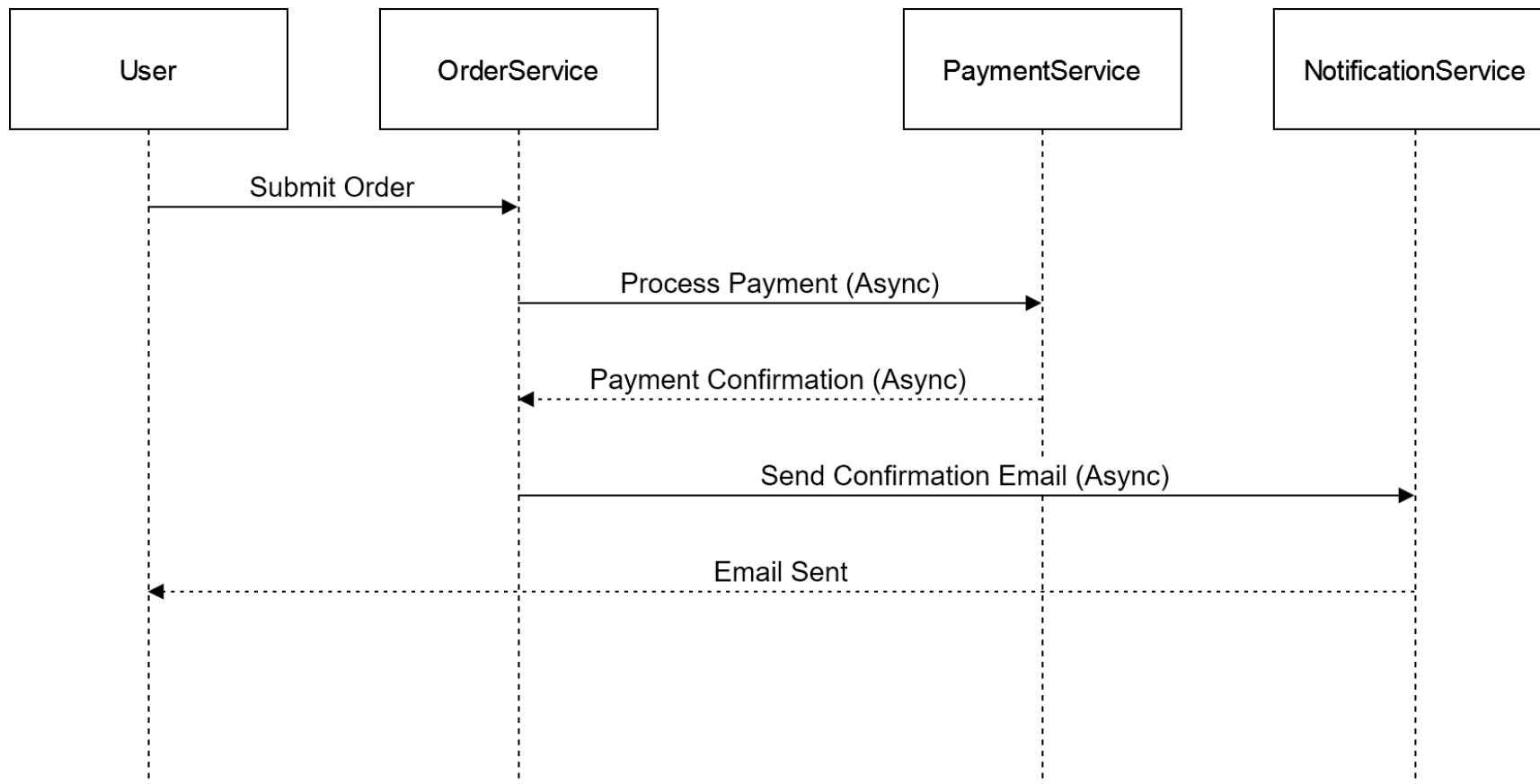
2.2 Architecture Views – Logical View

Represents the system's functional components and their relationships. It focuses on how the software is structured to handle the business requirements.



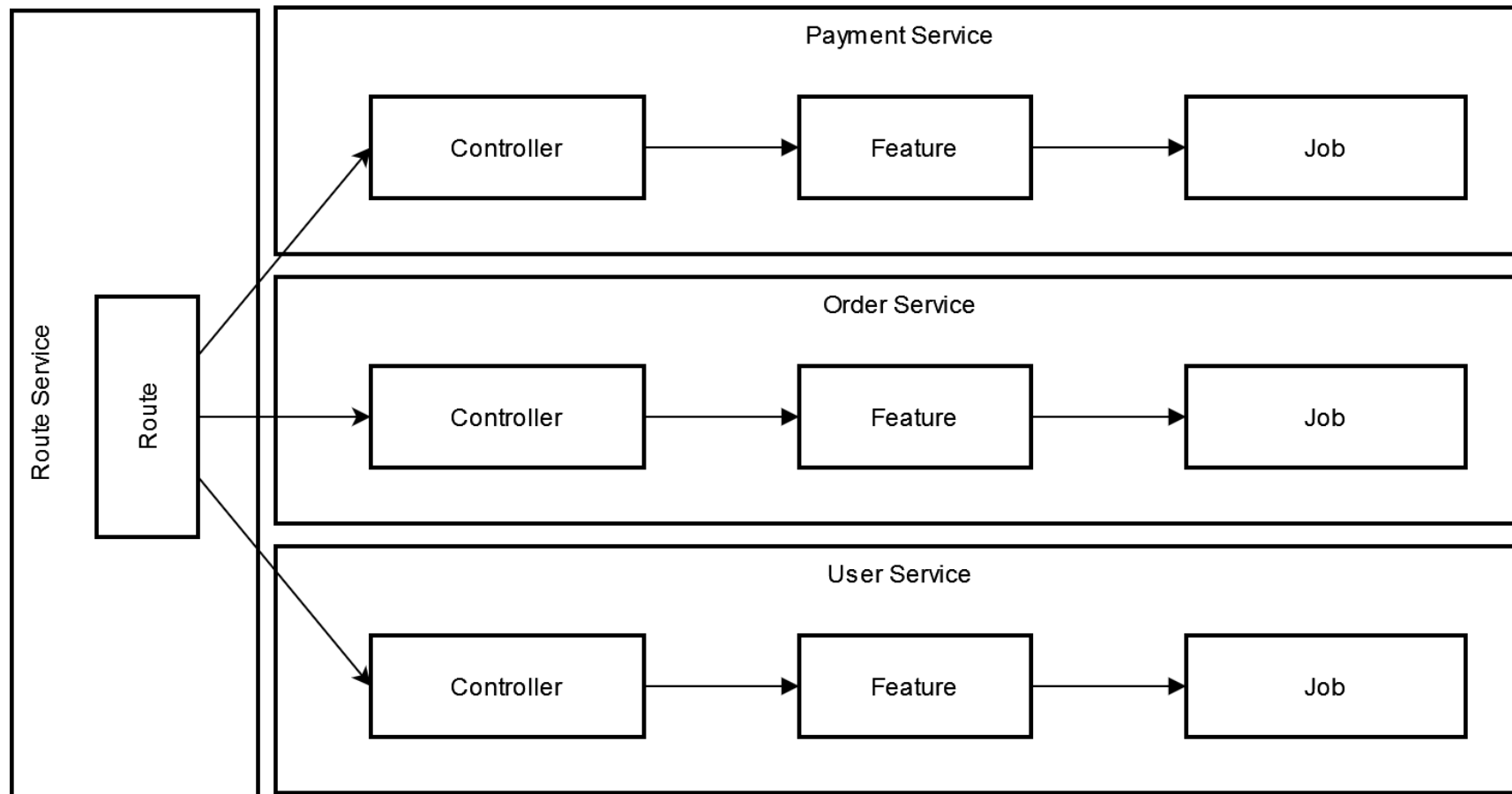
2.3 Architecture Views – Process View

Focuses on the runtime behavior of the system, particularly aspects like concurrency, synchronization, and interaction between processes.



2.4 Architecture Views – Development View

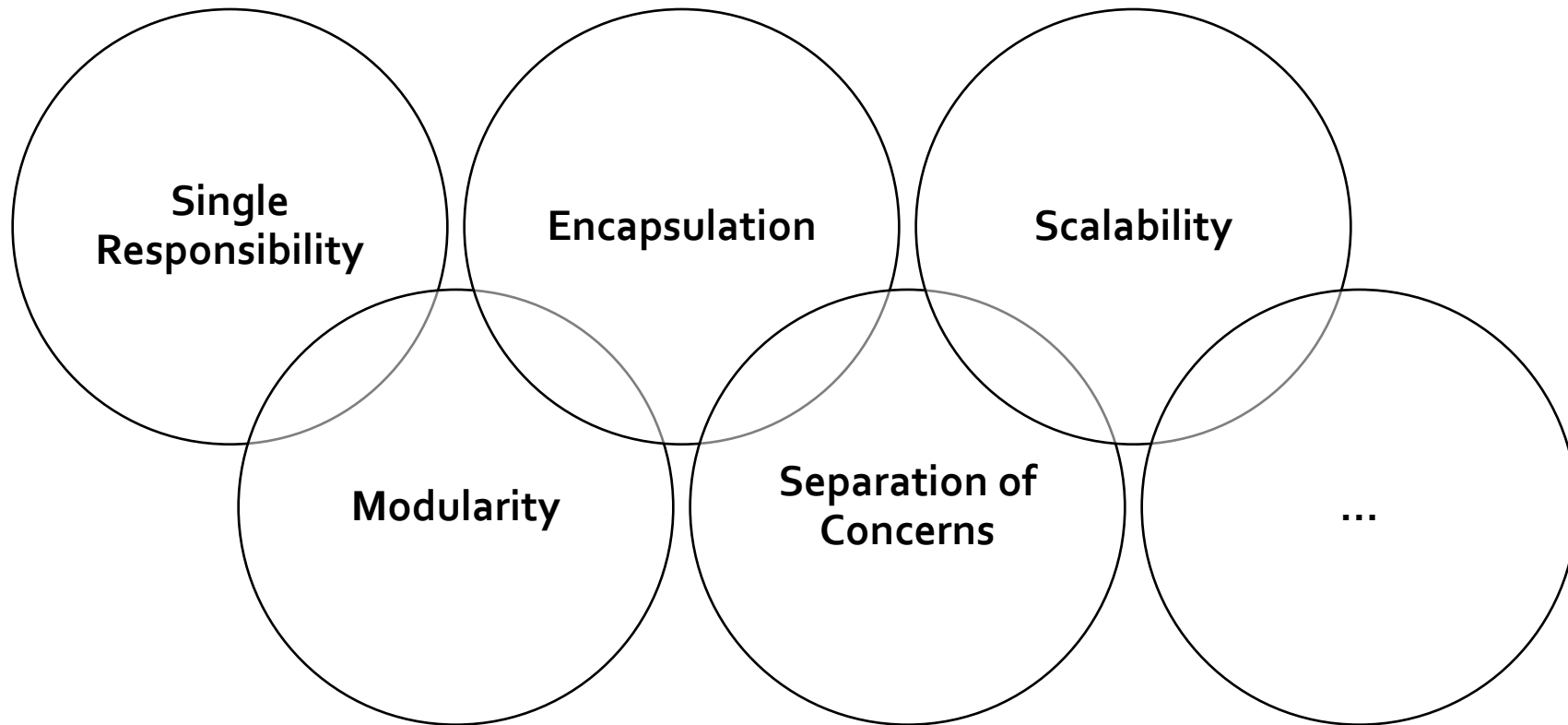
Describes the system from the perspective of the developer. It shows the organization of the code, modules, and packages.



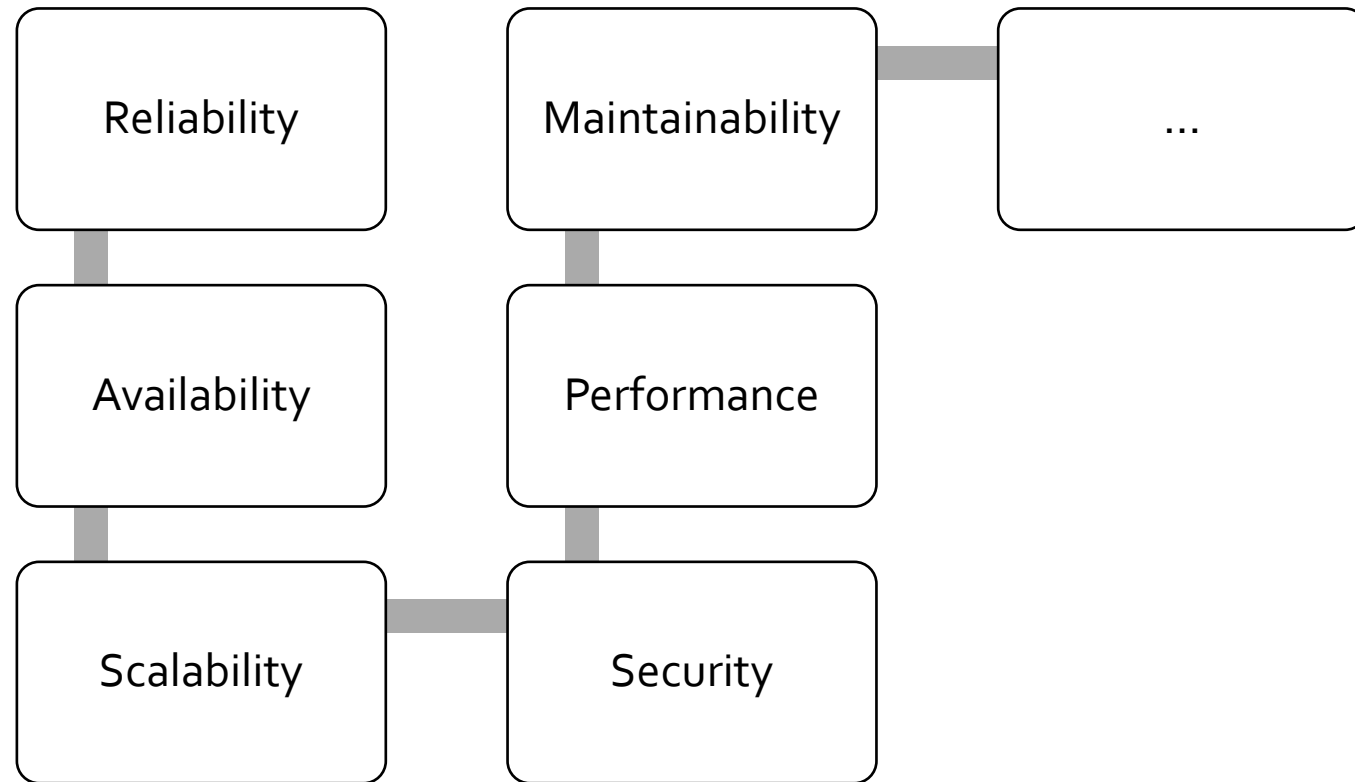
3. Architecture Patterns

Microservices	Monolithic	Serverless	Event-Driven
Service-Oriented	DDD	Clean	Plugin-Based
Hexagonal	Model-View-Controller	VIP	MVVM
VIPER	Client Server	Service Oriented	...

4. Architecture Principles



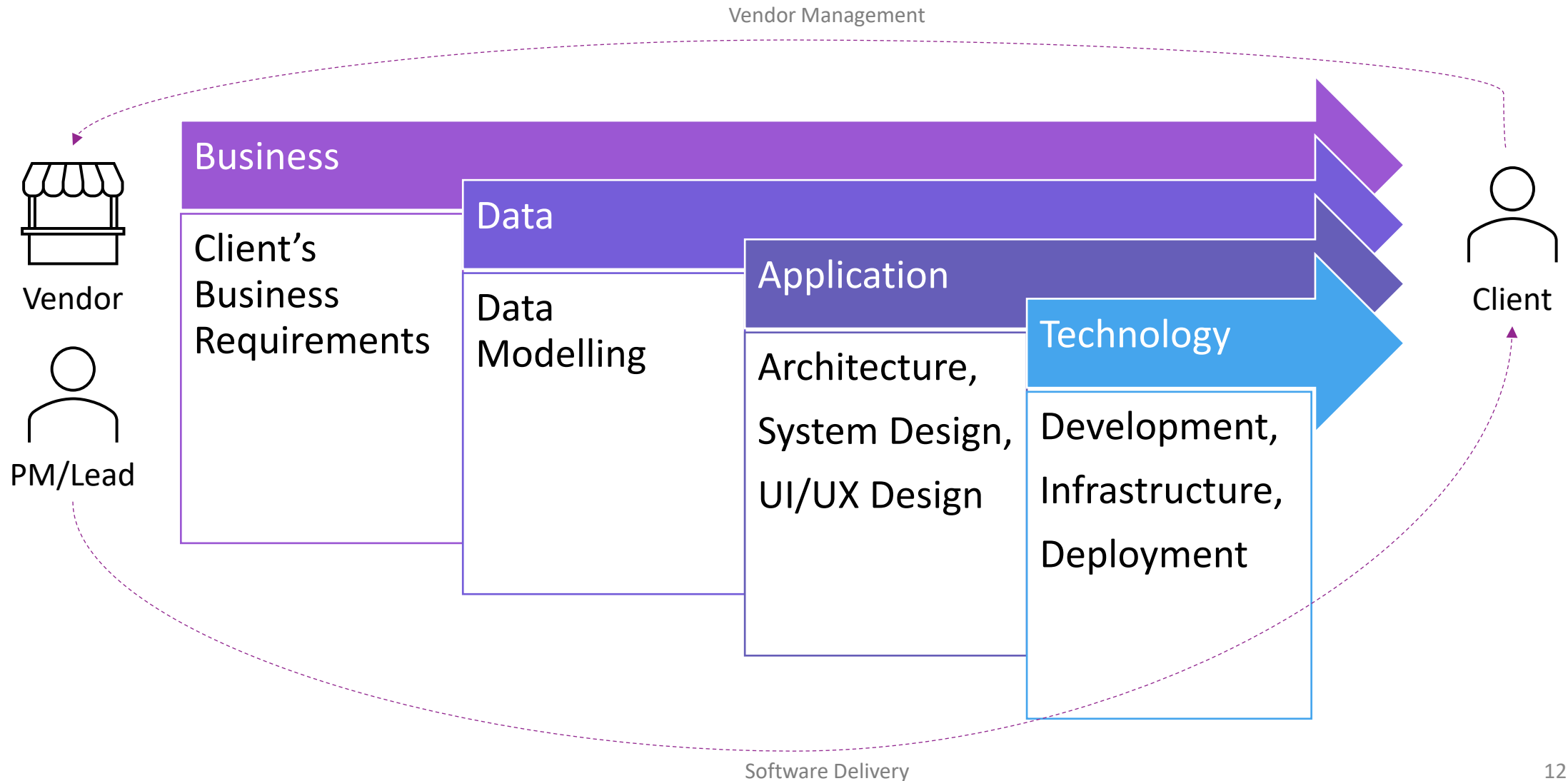
5. Architecture Quality Attributes



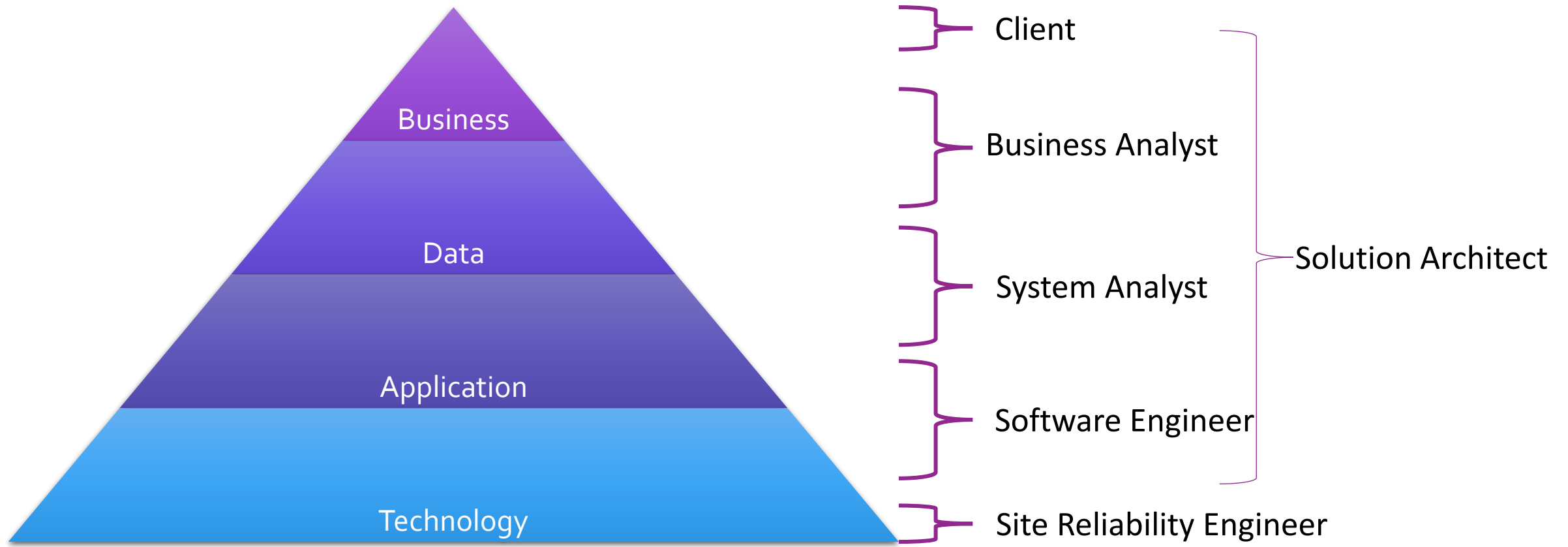
Implementation & Delivery

From building the software to deploying it into a production environment, where it becomes operational for end-users.

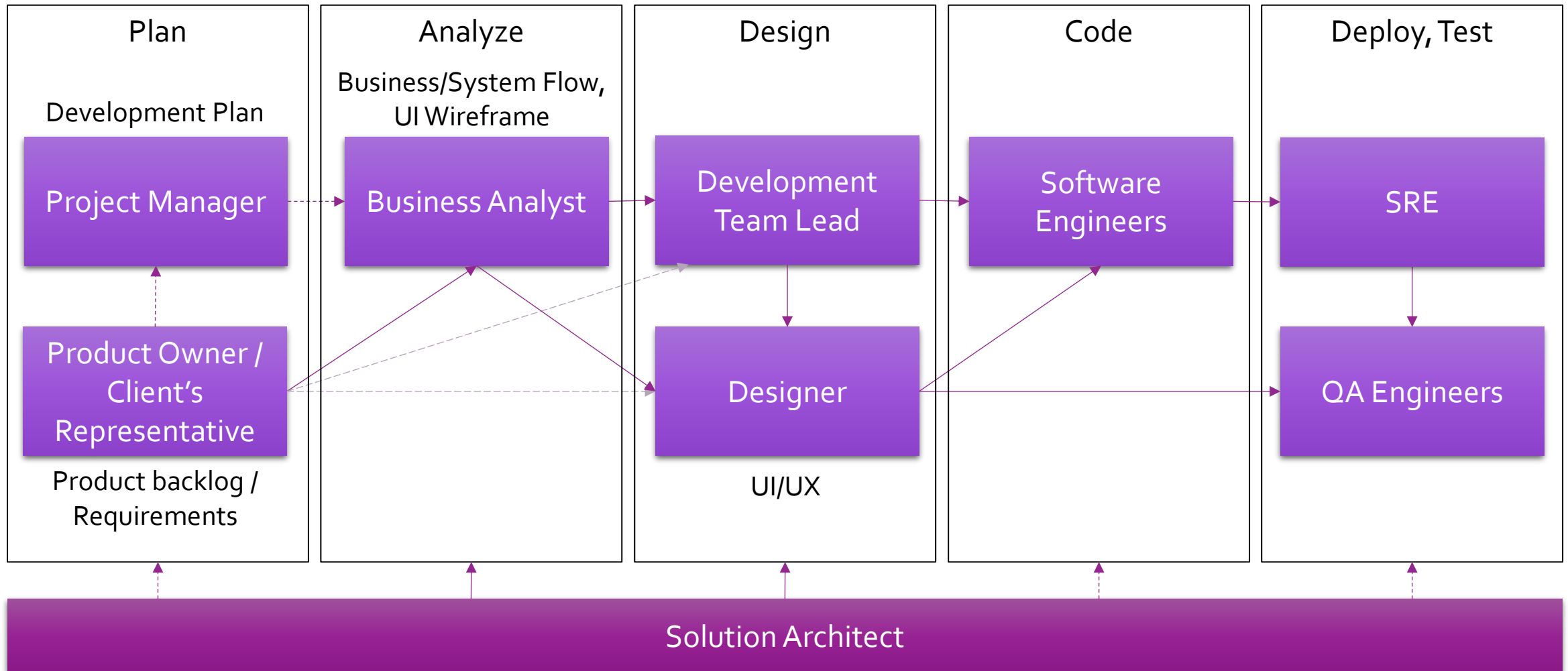
Domains and Process



Domains and Roles



Workflow and SDLC



Software Development Activities

#	Activity	Role	Output	Document Tools
1	Requirement Analysis	BA	Diagrams, Wireframe, BRS	whimsical, Word
2	Conceptual Data Modeling	SA	ER Diagram	drawio
3	UI Design & Prototyping	Designer	Design Mockup, Prototype	Figma, drawio, whimsical
4	Logical & Physical Data Modeling	SA	DB Schema	dbdiagram, dbdocs
5	System Design	SA	System design artifacts	drawio, whimsical
6	Prototype Usability Testing	BA, QA	Prototype, Test Plan/Case	figma
7	API Definition	SA / Lead	API Documentation	Postman
8	Coding	SE	Software	VSCode, .md files
9	Testing & Creating User Manual	QA	Issues, User Guide Docs	Excel, Words, Clickup

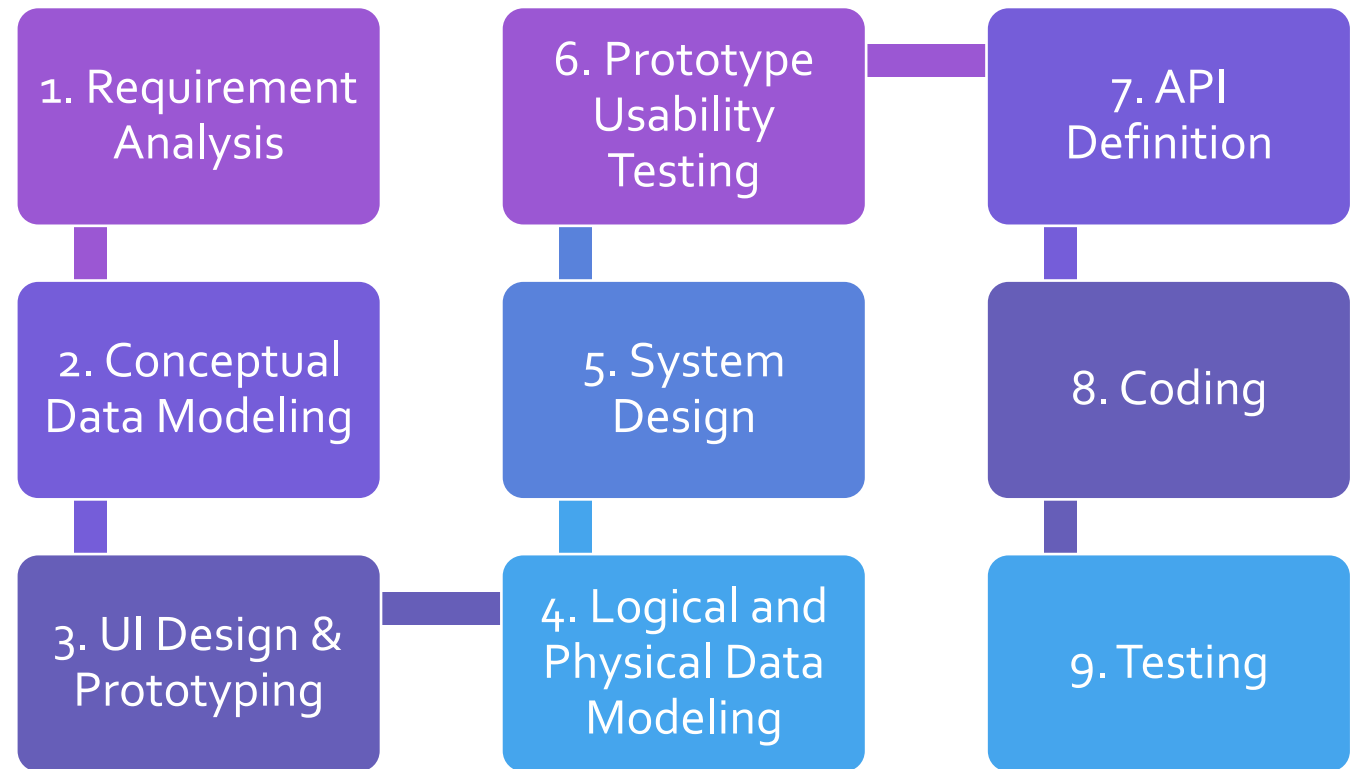
Activities and Domains

#	Activity	Architecture Domain(s)	Description
1	Requirement Analysis	Business	Understanding business goals, processes, and needs.
2	Conceptual Data Modeling	Data	High-level representation of business data relationships.
3	UI Design & Prototyping	Application	Interface design and user experience for the application.
4	Logical & Physical Data Modeling	Data	Schema design and data structure implementation.
5	System Design	Application	System architecture, modular breakdown, and behavior.
6	Prototype Usability Testing	Application	Validation of usability and interaction design.
7	API Definition	Application + Technology	Interface between system components and external access (technical + app logic).
8	Coding	Technology + Application	Writing software logic and managing infrastructure interactions.
9	Testing & Creating User Manual	Application + Technology	Verifying correctness, performance, and functional reliability.

Development Process

Order of activities might be adjusted according to the situation. That's about your **Wisdom**.

How wise are you?



Case Study Document

Creating and Executing Implementation for Web-Based POS Software

Defined Development Activities

1. Requirement Analysis
2. Data Modeling
3. UI Design & Prototyping
4. System & API Design
5. Coding & Development
6. UAT Deployment, Testing & Usability Feedback
7. Production Deployment, User Manual & Training

1. Requirement Analysis

- Responsible Role: Business Analyst (BA)
- Objective: Understand the sales flow, inventory control, and basic reporting requirements.
- Activities:
 - › Interview stakeholders (e.g., store owners)
 - › Define core features: sales entry, product catalog, user login, daily sales report
 - › Prepare a Business Requirement Specification (BRS)
- Output: BRS Document or Draft Notes
- Tools: Whimsical, Word

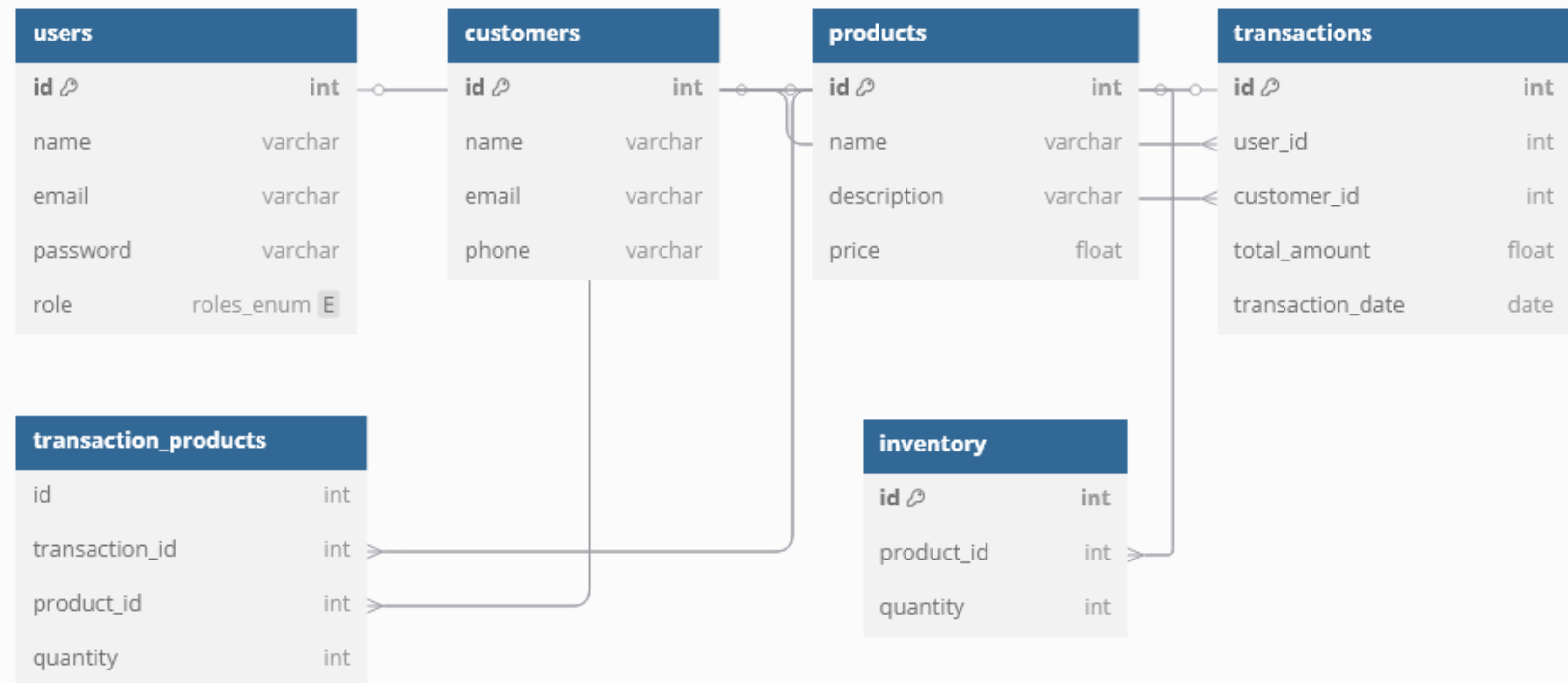
Requirements – Draft Notes

- Functional
 - › Sales Transactions: Enter product details, calculate total, update inventory, generate receipts.
 - › Product Management: Add, update, and delete products with name, description, price, and quantity.
 - › User Management: Role-based authentication (Cashier, Manager, Admin) with secure access control.
 - › Reporting: Basic sales, inventory, and transaction history reports.
- Non-Functional
 - › Usability: Intuitive interface requiring minimal training for store employees.
 - › Availability: Continuous system operation during store hours with minimal interruptions.
 - › Performance: Real-time transaction processing and fast report generation.
- Business Data Collection
 - › Master Data: users, products, customers (spreadsheet or papers)
 - › Transaction Data: sales transactions (paper invoices or written note documents)
- Business Process Analysis

2. Data Modeling (Merged Step)

- Responsible Role: System Analyst (SA)
- Objective: Design database structure for products, users, sales, and inventory.
- Activities:
 - › Create ER Diagram
 - › Define logical schema and DB tables
- Output: DB Schema, ER Diagram
- Tools: dbdiagram

Database Design



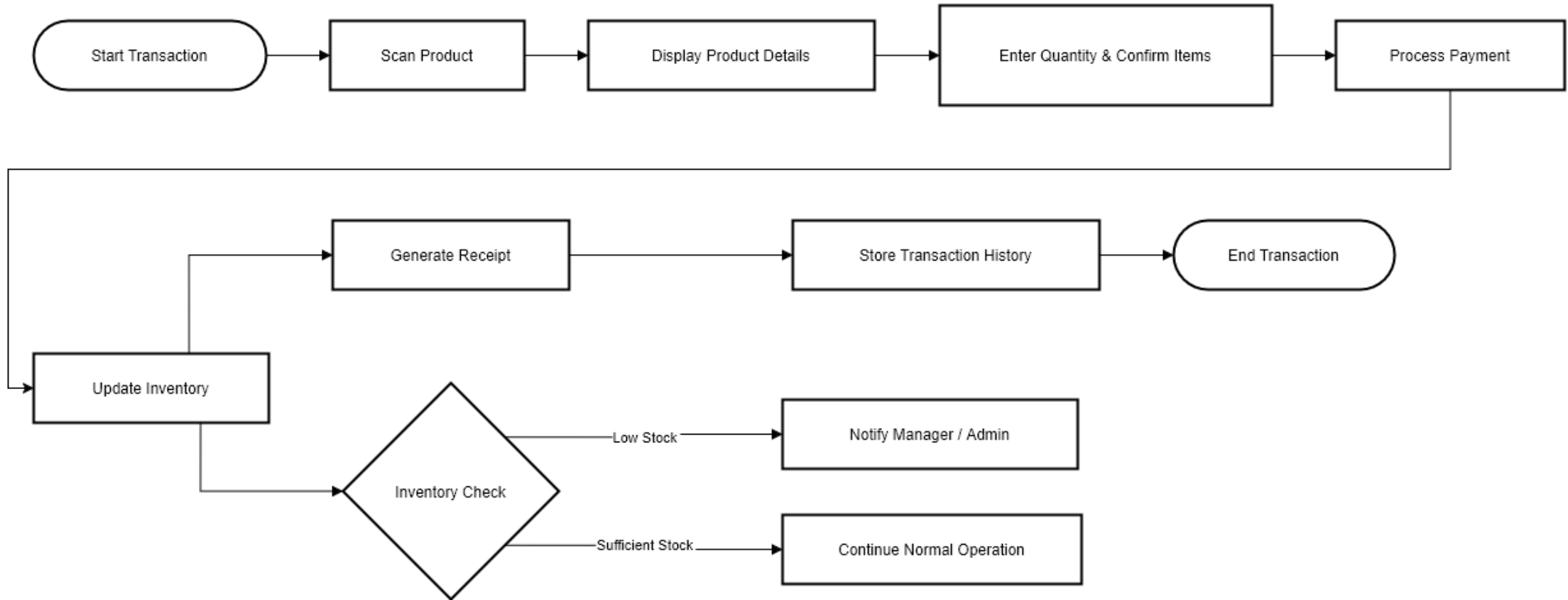
3. UI Design & Prototyping

- Responsible Role: Designer
- Objective: Design simple, intuitive screens for POS operations.
- Activities:
 - › Sketch screen layouts for login, sales entry, product listing, reports
 - › Build interactive mockups
- Output: Design mockups, wireframes
- Tools: Figma, Whimsical

4. System & API Design (Merged Step)

- Responsible Role: Solution Architect / SA
- Objective: Define system architecture and backend API contracts.
- Activities:
 - › Identify modular services: Auth, Product, Sales
 - › Define RESTful endpoints
 - › Prepare system architecture diagram
- Output: System Design Artifacts, API Docs
- Tools: Draw.io, Postman

System Design – Sale Transaction



API Design

Endpoint	Method	Description	Request	Response
/api/login	POST	Logs a user in and returns a JWT token.	{ "email": "user@example.com", "password": "password" }	{ "token": "jwt_token_here" }
/api/users	GET	Retrieves a list of users (Admin only).		[{ "user_id": 1, "name": "John Doe", "role": "Cashier" }, ...]
/api/users	POST	Creates a new user (Admin only).	{ "name": "John Doe", "email": "john@example.com", "password": "password", "role": "Cashier" }	{ "user_id": 123, "name": "John Doe", "role": "Cashier" }
/api/products	GET	Retrieves a list of products.	N/A	[{ "product_id": 1, "name": "Product A", "price": 100 }, ...]
/api/products	POST	Creates a new product.	{ "name": "Product Name", "price": 100, "description": "Description" }	{ "product_id": 1, "name": "Product Name", "price": 100 }
/api/transactions	POST	Creates a new transaction (Sales).	{ "user_id": 1, "customer_id": 1, "products": [{ "product_id": 1, "quantity": 2 }, ...] }	{ "transaction_id": 12345, "total_amount": 300, "status": "Success" }
/api/inventory	GET	Retrieves the current inventory for each product.	N/A	[{ "product_id": 1, "quantity": 100 }, ...]
/api/inventory	POST	Updates product stock levels.	{ "product_id": 1, "quantity": 100 }	{ "product_id": 1, "new_quantity": 100 }

5. Coding & Development

- Responsible Role: Software Engineers (SE)
- Objective: Implement frontend and backend of POS system.
- Activities:
 - › Build using Laravel (Backend) and Vue.js (Frontend)
 - › Implement API routes and database models
- Output: Working Software
- Tools: VSCode, Git, Swagger UI (for auto API docs)

6. UAT Deployment, Testing & Usability Feedback

- Responsible Role: DevOps/Cloud Engineer, QA, BA
- Objective: Ensure system functions as expected and meets business needs.
- Activities:
 - › Manual testing of features
 - › Feedback loop with store users
- Output: Issue list, Test Cases, User Feedback
- Tools: Excel, Clickup

7. Production Deployment, User Manual & Training

- Responsible Role: DevOps/Cloud Engineer, QA, BA
- Objective: Final deployment and end-user documentation.
- Activities:
 - › Deploy on AWS or DigitalOcean
 - › Prepare a simple guide for cashier/staff
- Output: Deployed system, User Guide
- Tools: Word, PDF, Clickup



[Aung Kyaw Minn](#)

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